Circuit Breaker Condition Assessment by Vibration and Trip Coil Analysis

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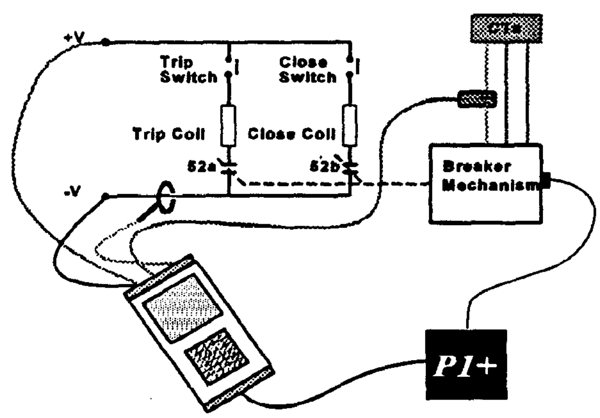
Introduction

In todays electricity distribution industry, fixed or shrinking maintenance budgets and the requirements to maintain or improve plant reliability, and thus reduce customer outages, are creating conflicting pressures for maintenance managers. Extending maintenance intervals, or cancelling maintenance to meet budget constraints can only be viewed as a short term solution, that, at best, will last only a few years.

The traditional approach of a scheduled maintenance programme is time consuming and resource hungry. Routine maintenance is therefore not a feasible option in todays industry. In addition to this, current thought is that scheduled maintenance does not guarantee plant reliability, indeed there is growing evidence that many circuit breaker faults may only manifest themselves several weeks after maintenance, or may even be caused by the maintenance itself! The well known expression - "if it ain't broke, don't fix it", is therefore becoming part of the maintenance philosophy in todays industry.

If, however, traditional maintenance programmes are to be abandoned, there needs to be an alternative approach for ensuring plant reliability. Clearly, some method of assessing a circuit breaker's condition or readiness to trip is required. This, due to the considerations outlined above, must be a quick, non-invasive test which enables the operator to know how the breaker would operate from 'cold'. This method of screening plant would enable maintenance money and resources to be targeted in a much more effective manner.

This paper introduces the Profile System, which has been developed by Kelman Ltd. in order to address these issues. This instrument provides a quick and easy method of assessing the condition of circuit breakers by simply performing an open and close operation. The two main techniques used by this instrument to assess circuit breaker condition are trip coil profiling and vibration monitoring. These methods are outlined in the following sections.

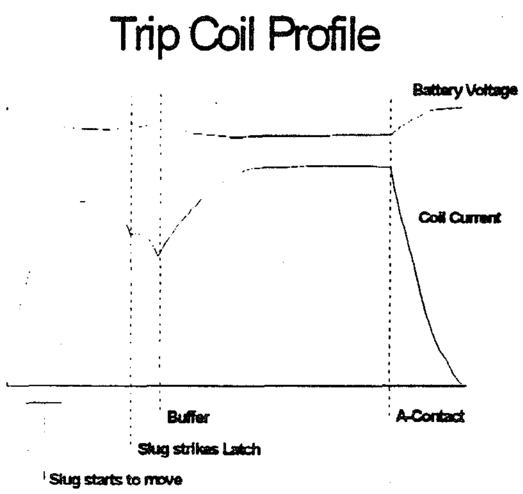


Trip Coil Profiling

Trip coil profiling seeks to gain an insight into the condition of the circuit breaker by means of examining the 'profile' of the current which flows through the trip/close coils during circuit breaker operations.

The trip coil is an electromagnetic actuator which when energised causes an iron slug to strike and release the trip latch and release the spring. This coil can also be viewed as a sensor, in that, when an piece of magnetic material is pushed through a coil, a voltage is induced in that coil. It can be seen therefore, that while current flowing through the coil effects a force upon the slug, the movement of the slug through the coil generates an e.m.f. in the coil, which in turn has an effect upon the current flowing through it.

Fig. 2 shows a typical current profile resulting from a trip coil operation. When the coil is first energised, the current through the coil increases exponentially until the force on the iron slug due to the current flowing through the coil exceeds both gravity and stiction. The slug then accelerates through the coil and the back e.m.f. produced can actually reduce the current flowing through the coil. The slug continues to move until it strikes the trip latch. Conservation of momentum at this point dictates a step drop in velocity of the slug, and this can be seen as a 'corner' or 'discontinuity' in the coil current profile. The combined mass of slug and latch continues to move until the trip latch releases the spring, which operates the main mechanism. Eventually, no further movement is possible, and the current rises to its maximum level, which is limited by the resistance of the coil. After the main contacts open, the auxiliary contacts operate, and the current through the trip coil is cut. The slug falls down back through the coil, and finally comes to rest at its starting position.



As can be seen from fig. 2, from the trip coil profile, the movement and timing of the slug can be traced as it moves from rest, hits the latch, and after releasing the latch, hits the buffer or end-stop. In addition, the time when the auxiliary contacts operate to cut the coil current will give an indication of the breaker operating time since the auxiliary contacts will typically be connected to the main mechanism, and usually operate after the main contacts have opened.

Each good operation of a breaker will produce the same current profile. Changes in the operating speeds or times of the various parts of the breaker are clearly evident when comparing a coil profile with the good trace. Poor initiation contacts, resistance to slug movement, delay in the tripping of the latch, faults in the trip coil, delay in, or non-operation of the main contacts, and bad, or poorly aligned auxiliary contacts will, and have all been seen to, alter the shape of the current profile.

Current profiles of all trip coils exhibit these main features even though the overall shape of the current trace may differ. There will, however be a common profile for each type of breaker and battery voltage configuration, so that it is not necessary to have a profile 'fingerprint' for each individual breaker in order to assess condition. Indeed, it has been found that with a little experience in interpreting coil current profiles, it is possible to diagnose problems in previously untested breakers.

The PROFILE uses a split-core d.c. current clamp to perform a test. This approach provides an easy method of capturing trip coil profiles. Being non-invasive, it also enables the breaker to be tested when on-line and therefore capture the first trip from 'cold'. This operation is the most critical from the aspect of the circuit breakers condition, as it is this operation which will clear any fault condition should the breaker be called upon to operate. In addition, this first operation will show up any stickiness or sluggishness in the operation of the circuit breaker, which will not usually be evident in subsequent operations.

D.C. Voltage Monitoring.

The condition of the substation battery can have as much effect on the operation of the circuit breaker as the condition of the breaker itself. The condition of the battery is not only important from the aspect of the breakers operation, but it is important for the operation of the entire substation. A faulty or weak battery will compromise the operation of the entire substation.

Included in the PROFILE is the facility to monitor battery voltage during a test. This not only shows the voltage of the battery before and after the operation, but also monitors the condition of the battery under load conditions. This measurement will also highlight any weaknesses in the wiring from the battery to the circuit breaker.

Connections to the battery supply are made using clip-on probes. This provides a quick and easy way to enhance the value of a trip coil test by also monitoring the state of the station battery.

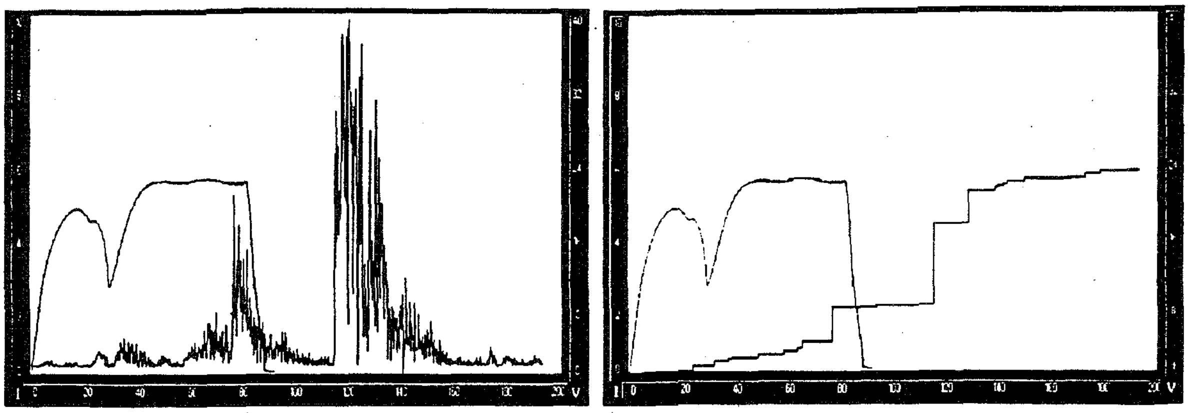
Main Contact Monitoring

The PROFILE also has an input for an a.c. current clamp which is clipped over the protection C.T. secondary wiring. This sensor enables the instrument to measure the time taken to for the breaker current to be cut. This sensor is also used to provide timings for close operations, giving the time to when the contacts first come together.

This test builds upon the previous two measurements by giving an insight into the operation of the main mechanism.

Vibration Analysis

All circuit breaker mechanisms are driven by a source of stored mechanical energy. The controlled release of this energy is used to drive the contacts open and closed. These operations result in characteristic vibration patterns in the breaker mechanism and interrupter assemblies. The details of the vibration are governed by the design of the circuit breaker, but it has been demonstrated by PNEM [Ref 1.] that the vibration signal generated during a circuit breaker operation contains certain features which allow the operation to be analyzed in great detail and in a way not possible using traditional techniques.



Kelman and PNEM have created a vibration analysis tool which has been incorporated into the PROFILE range. This makes vibration analysis easy to perform in the field.

The vibration signal is recorded by an accelerometer, which is magnetically attached to a suitable point on the circuit breaker structure. As the breaker is operated, the vibration signal is rectified and recorded. The recorded waveform is analyzed to pick out the commencement of mechanical events within the operation of the circuit breaker. These events are compiled to produce a step waveform, the size of each step corresponding to the magnitude of the vibrational energy in each event. This step waveform greatly simplifies the comparison of the sampled vibration data, and when viewed together with the raw vibration signal, eases the identification of the key points of the waveform..

The resulting traces, or vibration signature, are incorporated with the 'standard' profile features - trip coil current, d.c. voltage, and main contact operation - to give comprehensive picture of each circuit breaker operation.

Field testing has shown that the vibration fingerprint is every bit as repeatable as existing profile data, and that each circuit breaker design has its own distinct signature. Problems in the main breaker mechanism such as misaligned contacts or dashpot mal-operation will alter the fingerprint obtained.

The vibration analysis test is non-invasive, and provides the user with information on the circuit breaker operation beyond the simple electrical operation of latch and main contacts. Coupling this test with the trip coil profiling test provides a picture of the condition of the entire breaker.

Impact on Working Methods

Testing circuit breakers with the PROFILE system not only provides much more information on the breakers condition than was previously possible, but at the same time reduces the time taken for each test. A test can be carried out in about five to ten minutes simply by opening and closing the breaker. Given the same resources, a circuit breaker can be tested more frequently.

The breaker does not need to be taken out of service in order to perform the test, thus enabling the first trip to be captured. The non-invasive nature of the test also has benefits. If a circuit breaker is found to be fully operational, there will be no danger of the testing itself introducing new faults.

The results obtained provides an effective indication of the circuit breaker. Faults can be classified in levels of seriousness, so that maintenance effort can be scheduled and targeted where it is most required.

Test Results

Kelman have been involved with several customers in testing the techniques described in this paper. Results obtained for both trip coil profiling and vibration analysis have shown that both approaches produce repeatable results. Results from tests comparing normal operations with faulty operations for different types of circuit breaker will be the subject of another paper.

Conclusions

Non-invasive methods of monitoring the condition of a circuit breaker have been presented. With the simplest configuration, trip coil profiling provides information on the majority of the breaker's operation, and has already become an accepted method of testing. Vibration analysis permits the user to extend the test to cover the operation of the main breaker mechanism, and thus provide a complete picture of the state of the circuit breaker.

References

1. "PNEM ontwikkelt 'Fingerprint' voor MS schakelaars" Koen Krikke, Jan Peters, Michiel Wilde. Energietechniek, Oktober 1996.
2. "Condition Monitoring Techniques can enhance the Performance of Electricity Networks". Robin Watson, D. Mehrdad Rezazadeh, Koen Krikke, Michiel Wilde. DADSM October 1996.